Experiment 29: Evaluation Metrics for Model Performance

Aim: To evaluate a machine learning model using accuracy, precision, recall and F1-score.

Algorithm:

1. Import the required libraries.
2. Load the Iris dataset.
3. Split data into training and testing sets.
4. Train a Decision Tree classifier.
5. Calculate evaluation metrics.
6. Display the results.

Code:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

iris = load_iris()
X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = DecisionTreeClassifier()
model.fit(X_train, y_train)

pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, pred))
print("Precision:", precision_score(y_test, pred, average="weighted"))
```

print("Recall:", recall_score(y_test, pred, average="weighted"))

print("F1 Score:", f1_score(y_test, pred, average="weighted"))

Input:

Dataset: Iris dataset

Test size: 20%

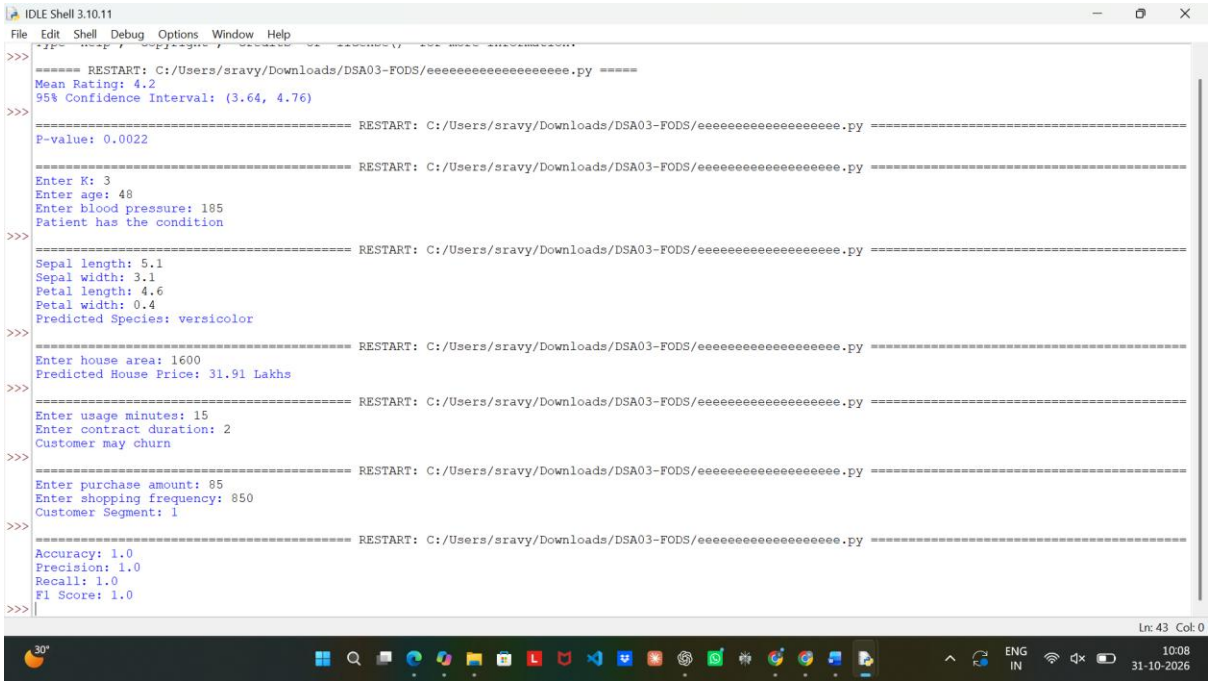
Output:

Accuracy: 1.0

Precision: 1.0

Recall: 1.0

F1 Score: 1.0



Result: The model performance was successfully evaluated using four metrics.